What If Attention Feeds Back Into Itself?

Self Referring Attention

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Abstract

What if attention feeds back into itself? It becomes a loop.

We define a mathematical object whose input is its own output, with an external input. We prove that five of the components are forced: the accumulation scaling, capture threshold, blend-rate boundaries, scoring mechanism, and feedback readout. There are four symmetrical invariants: the blend rate, the scoring rule, the query map, and the scale of capture. The simplex map is boundedly invariant within a defined hierarchy ($\mathcal{A} \supset \mathcal{A}_+ \supset \{\exp\}$). The qualitative dynamics hold for all domains with zero numerical parameters.

The loop shifts what it pays attention to in the world based on its past reactions. It has a strict carrying capacity $|M^*|$, which retrieval degrades as its capacity fills. A Lyapunov contraction consolidates impression vectors during the loop to effectively reduce the number of items, but if the capcity exceeds $|M^*|$ the loop permantely degenerates and consolidation cannot save it. The loop can calculate the retrieval degradation signal, but cannot autonomously disable capture. As a result, a structural asymmetry exists in which an external controller, of any form and separate from the loop, must read the signal and periodically disable capture, then reenable it once consolidation is complete, to prevent permanent degeneracy.

In an uncountably infinite observation space X, unique dynamics arise in which the loop is forced to concentrate on a limited set of items, irreversibly forget, and have path dependence, which do not arise in a finite environment. All the finite X proofs are computationally verified with Lean4 under standard axioms.

1 Introduction

We define the minimum loop in which a system's attention feeds back into itself. The system's past states and reactions shape what it pays attention to next.

The loop consists of five operations: observe, accumulate, retrieve, consolidate, and capture. Removing any operation breaks the system. The system observes its state, accumulates weighted sums, retrieves based on what's most familiar for the observation, consolidates similar impressions, and captures new observations. The outputs are then fed into the system as its own inputs.

The loop is structurally invariant in 4 dimensions with an infinite family of admissible choices. Additionally, the simplex map is boundedly invariant within a defined hierarchy ($\mathcal{A} \supset \mathcal{A}_+ \supset \{\exp\}$). The remaining choices of inputs and domains do not impact the qualitative dynamics of the system.

1. λ-*invariance*: any admissible blend rate in $\mathcal{L}$ produces equivalent qualitative dynamics,
2. h-*invariance*: any admissible scoring rule operates,
3. F-*invariance*: any R-bounded linear query map applies,
4. *scale invariance of capture*: query scaling by $\beta > 0$ leaves the structure unchanged.

5. *φ-invariance*: $\mathcal{A} \supset \mathcal{A}_+ \supset \{\exp\}$ classifies every result by the invariance strength required for each theorem by positivity and exponential normalization requirements.

Every proof in the finite-domain case has been formally verified in Lean 4 under standard axioms.

Structure. Section 2 defines the loop. Section 3 proves the forced and invariant classifications. Section 4 establishes the invariance hierarchy and its scope. Section 5 proves the core dynamical lemmas. Section 6 presents the carrying capacity and lifecycle results. Section 7 develops the support dynamics that arise when X is infinite. Section 8 collects open problems. The appendices provide accumulation rigidity details and formal verification.

2 Definitions

- Let (X, Σ, μ) be a σ-finite measure space, $V = \mathbb{R}^d$
- $S\colon X \to \mathbb{R}$: bounded measurable scoring function
- $E\colon X \to V$: measurable embedding with $R := \sup_{\tau \in X} \|E(\tau)\| < \infty$
- $K\colon V \to V$: linear map
- *simplex map* φ: sends each bounded measurable $f\colon X \to \mathbb{R}$ to a probability measure on (X, Σ) absolutely continuous with respect to μ; write $a = d\varphi(f)/d\mu$ for its density, so $\int_X a\, d\mu = 1$
- We require φ to be *density-bounded*: for every $B > 0$ there exists $A(B) > 0$ such that $\|f\|_\infty \leq B$ implies $\|a\|_\infty \leq A(B)$
  - For exponential normalization with $\mu(X) < \infty$, $A(B) = \exp(2B)/\mu(X)$
  - For hardmax and sparsemax, $A(B)$ depends on the measure of the maximizer set
  - This condition is automatic when μ is counting measure (then $a(\tau) \leq 1$)
- *blend rate*: continuous function $\lambda\colon [0,1] \to [0,1]$ with $\lambda(0) = 1$ and $\lambda(1) = 0$
- Initial conditions: $M_0 = \emptyset$, $C_0(\tau) > 0$, $N_0(\tau) = 1$
- φ is *positive* if $a(\tau) > 0$ for μ-a.e. τ whenever f is bounded
- *exponential normalization*: $\varphi(f)(\tau) = \exp(f(\tau))/\int_X \exp(f)\, d\mu$ is positive when the integral converges (requires $\mu(X) < \infty$ or $f(\tau) \to -\infty$ sufficiently fast outside a finite-measure set). Hardmax and sparsemax are not positive.

The complete loop

(X, Σ, μ) σ-finite, $V = \mathbb{R}^d$, φ any simplex map, λ any blend rate, $E\colon X \to V$ measurable with $R = \sup_{\tau \in X} \|E(\tau)\| < \infty$, $K\colon V \to V$ linear.

$$M_0 = \emptyset,\ \ C_0(\tau) > 0,\ \ N_0(\tau) = 1.$$

①	$a = d\varphi(S + \sigma + b)/d\mu$	**observe**
②	$C \leftarrow C + a \cdot (t+1)/C$	**accumulate**
	$N \leftarrow N + 1$	
	$\sigma = C/N$	
③	$q = \int_X a\, E\, d\mu$	**retrieve**
	$w = \varphi(qM^\top)$	
	$\ell = wM$	
	$b(\tau) = E(\tau)^\top K \ell$	
④	$\lambda_j = \lambda(w_j / \max w)$	**consolidate**
	$M^* = \varphi\big((M + q)M^\top\big)M$	
	$M \leftarrow (1 - \lambda)M + \lambda\, M^*$	
⑤	$\theta = \max(qM^\top)$	**capture**
	$M \leftarrow M \cup \{E(\tau) : q^\top E(\tau) > \theta\}$	
①	$a = d\varphi(S + \sigma + b)/d\mu$	**observe**

The five steps form a deterministic dynamical system.

Finite-X version. When X is finite with counting measure, the loop simplifies to discrete sums; see Appendix A.

The rest of this section defines each component formally.

Definition 1 (Scoring)**.**

$$S'(\tau) = S(\tau) + \sigma(\tau) + b(\tau) \tag{1}$$

The base score is perturbed by two feedback terms: salience from the accumulator and a bias from retrieval. Additivity is the canonical choice; every theorem holds under any admissible scoring rule $S' = h(S, \sigma, b)$ (Proposition 18).

Definition 2 (Observation)**.**

$$a = \frac{d\varphi(S')}{d\mu}, \quad \int_X a\, d\mu = 1. \tag{2}$$

Definition 3 (Accumulation)**.**

$$C_{t+1}(\tau) = C_t(\tau) + a_t(\tau) \cdot \frac{t+1}{C_t(\tau)} \tag{3}$$

$$N_{t+1}(\tau) = N_t(\tau) + 1 \tag{4}$$

$$\sigma_{t+1}(\tau) = \frac{C_{t+1}(\tau)}{N_{t+1}(\tau)} \tag{5}$$

Definition 4 (Retrieval).

$$q = \int_X a(\tau)\, E(\tau)\, d\mu(\tau) \tag{6}$$
$$w = \varphi(q\, M^\top) \tag{7}$$
$$\ell(q, M) = wM \tag{8}$$
$$b(\tau) = E(\tau)^\top K\ell \tag{9}$$

The context vector q is the observation-weighted combination of embeddings (a Bochner integral; it converges since $\|aE\| \le R \int a\, d\mu = R$). The retrieval weights w score each stored impression's relevance. The bias $b\colon X \to \mathbb{R}$ is a measurable function.

Definition 5 (Consolidation).

$$\lambda_j = \lambda\left(\frac{w_j}{\max_{k\in M}\ w_k}\right) \tag{10}$$
$$T(M, q) = \varphi\big((M + q)\, M^\top\big)\, M \tag{11}$$
$$M' = (1 - \lambda)\, M + \lambda\, T(M, q) \tag{12}$$

Throughout this paper, $\lambda(r) = 1 - r$ unless otherwise noted. Actively retrieved impressions stay sharp ($\lambda_j = 1 - w_j/w_v \to 0$ as $w_j/w_v \to 1$); others drift toward their neighbors. The most-retrieved impression is preserved exactly ($\lambda_v = 0$).

Definition 6 (Capture). Adopting the convention $\max \emptyset = -\infty$:

$$M_{t+1} = M_t \cup \{E(\tau) : q^\top E(\tau) > \max_{k\in M_t} q^\top m_k\} \tag{13}$$

An element is admitted when its dot-product similarity to the current context exceeds that of every existing member.

Notation.

- $v = \arg\max_{k\in M} w_k$: the *anchor* (most-retrieved impression).
- $V(t) = \max_{j\neq v} \|m_j - v\|$: the *Lyapunov function* measuring state spread.
- $\gamma(t) = \min_{j\neq v} \lambda_j(1 - \sqrt{1 - \alpha_v^{(j)}})$: the *contraction rate*, where $\alpha_v^{(j)} = [\varphi((m_j + q)M^\top)]_v$.
- $\rho = \max_j w_j$: the *retrieval concentration* (sharpness of retrieval).
- $g = q^\top m_v - \max_{k\neq v} q^\top m_k$: the *score gap*.
- $R = \sup_{\tau\in X} \|E(\tau)\|$: the *embedding radius* (finite by assumption).
- $|M^*| = 1 + \frac{1-c}{c}\exp(2R^2)$: the *critical state size* for retrieval threshold c.

3 Rigidity

We now show that the loop's structure is almost entirely forced. The accumulation scaling, capture threshold, blend-rate boundary conditions, scoring mechanism, and feedback readout are each uniquely determined by consistency requirements. No alternatives exist within the class of parameter-free maps. The remaining components are either invariant (Section 4) or instance-defining. Given any instance, zero numerical parameters remain.

Proposition 7 (Rigidity of the accumulation rate; $\mathcal{A}$). *Replace $t + 1$ with an arbitrary scaling $f(t)$ in the accumulation rule: $C_{t+1} = C_t + a_t \cdot f(t)/C_t$. For $\sigma = C/N$ to converge to a nonzero finite constant under persistent observation ($a \ge \delta > 0$), $f(t) \sim t$ is necessary.*

Proof. Since $N = N_0 + t$, convergence of $\sigma = C/N$ to a nonzero constant requires $C = \Theta(t)$. The increment satisfies $C_{t+1}^2 - C_t^2 = 2a_t f(t) + a_t^2 f(t)^2/C_t^2 \geq 2\delta\, f(t)$, so $C_t^2 \geq 2\delta \sum_{s=0}^{t-1} f(s)$. For $C = \Theta(t)$ we need $\sum_{s \leq t} f(s) = \Theta(t^2)$, which forces $f(t) = \Theta(t)$ under the convergence requirement (see Appendix A). If f grows slower (e.g. $f \equiv 1$), then $C = O(\sqrt{t})$ and $\sigma \to 0$. If f grows faster (e.g. $f(t) = t^2$), then $C = \Omega(t^{3/2})$ and $\sigma \to \infty$. The unique power law is $f(t) = \Theta(t)$. □

Remark 8 (Non-degeneracy of the accumulation scaling). The rejected scalings are not alternative design choices. Both destroy the loop's ability to function. If f is sublinear, then $\sigma \to 0$ under persistent observation: the salience of an actively attended element decays to zero, so the element becomes invisible to the observation mechanism. If f is superlinear (e.g. $f(t) \geq ct^2$), then $\sigma \to \infty$: a single element's salience dominates all others indefinitely, and φ concentrates the entire observation budget on that element. The linear scaling $f(t) \sim t$ is the unique case in which persistent observation stabilizes salience.

Lemma 9 (Idempotent capture; $\mathcal{A}$)**.** *For a fixed query q and fixed M, capture applied twice yields the same result as capture applied once.*

Proof. After capture, every admitted $E(\tau)$ satisfies $q^\top E(\tau) > \max_{k \in M} q^\top m_k$ and is now in M'. So $\max_{k \in M'} q^\top m_k \geq q^\top E(\tau)$ for all admitted elements. A second application admits nothing: previously admitted elements already belong to M', and previously rejected elements had $q^\top E(\tau) \leq \max_{k \in M} q^\top m_k < \max_{k \in M'} q^\top m_k$. □

Proposition 10 (Rigidity of the capture threshold; $\mathcal{A}$)**.** *Among parameter-free, always-defined, memoryless thresholds,* max *is the unique one consistent with idempotent capture (Lemma 9).*

Proof. Let $\theta(M, q)$ be any memoryless threshold. Idempotence (Lemma 9) requires that if $M' \supseteq M$ is the result of one capture pass, then a second pass with M' admits no new elements. This holds if and only if $\theta(M', q) \geq \theta(M, q)$ whenever $M' \supseteq M$, i.e. θ is non-decreasing under set enlargement.

Sufficiency of max: let $\theta(M, q) = \max_{k \in M} q^\top m_k$. If $M' = M \cup \{E(\tau)\}$ with $q^\top E(\tau) > \theta(M, q)$, then $\theta(M', q) = q^\top E(\tau) > \theta(M, q)$. More generally, $M' \supseteq M$ implies $\theta(M') \geq \theta(M)$, so idempotence holds.

Uniqueness: any function $\theta(M, q) = f(\{q^\top m_k : k \in M\})$ that is non-decreasing under set enlargement and depends only on the multiset of scores must satisfy $f(S \cup \{s\}) \geq f(S)$ for all S and all s. Among order statistics, the k-th largest is non-decreasing under enlargement for every fixed k, but it is undefined when $|M| < k$. Since $|M| = 1$ is a valid state, only $k = 1$ (the maximum) is always defined; any $k \geq 2$ additionally requires specifying k, a free parameter. Among non-order-statistic functions: the mean can decrease under enlargement (add an element with score below the mean), as can the median and any quantile. Any offset such as $\max - \epsilon$ or $\max + \epsilon$ introduces the parameter ϵ. Hence max is the unique threshold that is parameter-free, always defined, and non-decreasing under set enlargement. The set operation is union for the same reason: every alternative (replacement, deletion, pruning) requires either a parameter (which element to remove) or additional state (per-element retrieval counts, age tracking), so capture by union is the unique memoryless, parameter-free admission rule. □

Proposition 11 (Necessity of each step; $\mathcal{A}$; items (iv)–(v) use $\mathcal{A}_+$/exp. normalization results)**.** *Removing any single step from the loop causes at least one of the following to fail: boundedness ($\sigma = O(1)$), non-degeneracy ($a \geq \alpha > 0$), the feedback property (M influences observation), or the core consolidation dynamics.*

Proof. We verify each step by citing existing results.

(i) Without OBSERVE. The observation vector a is undefined; the query $q = aE$ (Definition 4), the accumulator update (Definition 3), and all downstream quantities are undefined. The loop is inoperative.

(ii) Without ACCUMULATE. The salience $\sigma = C/N$ does not exist. The scored input reduces to $S' = S + b$, eliminating the temporal record of persistent observation. Selective persistence (Theorem 21) fails: the system cannot distinguish persistently attended tokens from transient ones.

(iii) Without RETRIEVE. The bias $b = 0$ permanently. The feedback loop $\ell \to b \to a \to q \to w \to \ell$ is broken: M has no influence on observation. Consolidation blend rates $\lambda_j = \lambda(w_j/\max w)$ are undefined (w does not exist). The carrying capacity argument (Theorem 41) is vacuous.

(iv) Without CONSOLIDATE. By Lemma 26, each embedding vector enters M at most once and no row of M is ever modified. Retrieval degradation during waking is permanent and irreversible. The Lyapunov contraction (Lemma 32) is vacuous.

(v) Without CAPTURE. M remains at its initial value. If $M_0 = \emptyset$, retrieval is permanently undefined. If M_0 is nonempty, consolidation contracts all impressions to a single prototype (Corollary 37) and the system converges to a degenerate fixed point. The carrying capacity (Theorem 41) does not arise. □

Theorem 12 (Classification of the loop)**.** *Every component of the system falls into one of three categories.*

(a) Forced. *The accumulation scaling $f(t) \sim t$ (Proposition 7), the capture threshold* max *(Proposition 10), the blend-rate boundaries $\lambda(0) = 1$, $\lambda(1) = 0$ (Proposition 15), the dot-product scoring $s_j = q^\top m_j$ (Proposition 13), and the feedback readout $E^\top$ (Proposition 13) are uniquely determined by consistency requirements among linear, parameter-free maps. No alternative within this class exists.*

(b) Invariant. *The simplex map $\varphi \in \mathcal{A}$, the blend-rate interior $\lambda \in \mathcal{L}$, the query map $q = aF$ for any R-bounded F (Proposition 13(a)), the scoring rule $S' = h(S, \sigma, b)$ for any admissible h (Proposition 18), and the scale of the capture threshold (Corollary 19) are free choices, but the results do not depend on which admissible choice is made. Replacing one admissible φ, λ, F, h, or capture scale with another changes no theorem at its stated level.*

(c) Instance-defining. *The embedding E, linear map K (Proposition 13(d)), consolidation target T_j (Proposition 14), and the choice between soft and hard retrieval (Proposition 13(c)) specify which system one is studying. The admissible class for each is characterized by the referenced propositions.*

Each of the five steps is necessary (Proposition 11). The retrieval pipeline requires exactly four operations; no three-step pipeline suffices (Proposition 13). Given any instance, zero tunable numerical parameters remain: the dynamics are fully determined by the definitions.

Proof. (a) Propositions 7, 10, 15, and 13 each establish uniqueness within the class of parameter-free maps. (b) Proposition 15(b) establishes φ-invariance across the hierarchy $\mathcal{A} \supset \mathcal{A}_+ \supset \{\exp\}$; Proposition 13(a) establishes F-invariance; Proposition 18 establishes h-invariance; λ-invariance follows from the boundary-only dependence in Proposition 15(a); and Corollary 19 establishes capture-scale invariance. (c) Proposition 13(b)–(d) and Proposition 14 characterize the admissible class for each instance-defining component. Exhaustiveness: every symbol appearing in the loop definition (Section 2) is accounted for in (a), (b), or (c). Necessity of all five steps is Proposition 11; irreducibility of the pipeline is Proposition 13. □

Proposition 13 (Rigidity of the retrieval pipeline)**.** *The retrieval pipeline*

$$q = aE, \quad w = \varphi(qM^\top), \quad \ell = wM, \quad b = E^\top K\ell$$

is determined (up to the instance-defining parameter K) by the following requirements:

(a) The query map is invariant: for any measurable map $F\colon X \to V$ with $\|F(\tau)\| \le R$ for all τ, the query $q = \int a\,F\,d\mu$ satisfies $\|q\| \le R$; conversely, $\|\int a\,F\,d\mu\| \le R$ for all probability densities a forces $\|F(\tau)\| \le R$ μ-a.e. (by localizing a to shrinking neighborhoods of τ). Thus the class of admissible query maps is exactly the R-bounded measurable maps. E is the canonical zero-parameter representative. The bounded score gap $g \le 2R^2$ (Theorem 41(iii)) requires only $\|q\| \le R$.

(b) The dot-product scoring $s_j = q^\top m_j$ is the canonical parameter-free choice: since $m_j \in \operatorname{conv}(E(X))$ (by Lemma 25), the linear functional $q^\top(\cdot)$ attains its maximum at a vertex of $\operatorname{conv}(E(X))$. Nonlinear scoring may achieve its maximum in the interior. Among parameter-free bilinear forms, $q^\top m_j$ is the unique natural choice.

(c) $\ell = wM$ is instance-defining: hard retrieval $\ell = m_v$ preserves every theorem except quantitative bounds requiring smooth readout.

(d) E in the feedback $b(\tau) = E(\tau)^\top K\ell$ is forced as the unique parameter-free map $V \to L^2(X,\mu)$, namely the adjoint E^* of the embedding. K is instance-defining: $K = I$ preserves all theorems.

No pipeline with fewer than four operations produces the same phenomena: the chain

$$\mathcal{P}(X) \xrightarrow{q=\int aE\,d\mu} V \xrightarrow{w=\varphi(qM^\top)} \Delta^{|M|} \xrightarrow{\ell} V \xrightarrow{b=E^*K\ell} L^2(X,\mu)$$

passes through four distinct spaces, and removing any link eliminates retrieval weights, removes geometric feedback, or severs the state–observation link.

Proof. (a) The bounded score gap (Theorem 41(iii)) requires $g \le 2R^2$, which follows from $\|q\| \le R$ and $\|m_j\| \le R$. For any measurable map F with $\|F(\tau)\| \le R$ for all τ, the query $q = \int a(\tau)F(\tau)\,d\mu(\tau)$ satisfies $\|q\| \le R$ by convexity of the norm and $\int a\,d\mu = 1$. Conversely, if $\|\int a\,F\,d\mu\| \le R$ for every probability density a, then localizing a to shrinking neighborhoods of τ gives $\|F(\tau)\| \le R$ for μ-a.e. τ. Hence the admissible class is exactly the R-bounded measurable maps $\{F : \|F(\tau)\| \le R\ \mu\text{-a.e.}\}$. All downstream bounds ($\|q\| \le R$, $|b(\tau)| \le R^2\|K\|$, $q \in \overline{\operatorname{conv}}(F(X))$) hold for any such F, so the choice is invariant. E is the canonical representative that introduces no additional parameters.

(b) Since $m_j \in \operatorname{conv}(E(X))$ by Lemma 25, the linear functional $q^\top(\cdot)$ attains its maximum at an extreme point of $\operatorname{conv}(E(X))$, which is of the form $E(\tau)$. Nonlinear scoring (e.g. $-\|q - m_j\|^2$, concave in m_j, or $q^\top m_j/\|m_j\|$, rational) may achieve its maximum in the interior of $\operatorname{conv}(E(X))$ rather than at a vertex. Any bilinear form $q^\top W m_j$ with $W \ne I$ is equivalent to reparametrizing the embedding as $E' = WE$ when W is invertible. For singular W, $q^\top W m_j$ reduces to scoring in a lower-dimensional subspace, which is a degenerate case of the dot-product with a rank-deficient embedding. Thus $q^\top m_j$ uniquely represents the class of parameter-free bilinear forms up to reparametrization.

(c) The retrieval output ℓ appears in the system only through $b = EK\ell$. No theorem about the carrying capacity references ℓ directly. All other theorems are unchanged under hard retrieval $\ell = m_v$.

(d) Since $b\colon X \to \mathbb{R}$ and $\ell \in V$, a map $V \to L^2(X,\mu)$ is needed. The adjoint $E^*\colon v \mapsto [\tau \mapsto E(\tau)^\top v]$ is the unique parameter-free choice. Any alternative $F^* \ne E^*$ introduces a new parameter. The bounded-bias bound $\|b\|_\infty \le R^2\|K\|$ (Corollary 27) holds for any K. Setting $K = I$ preserves every theorem with $\|K\| = 1$.

(Irreducibility.) The chain $\mathcal{P}(X) \xrightarrow{q=\int aE\,d\mu} V \xrightarrow{w=\varphi(qM^\top)} \Delta^{|M|} \xrightarrow{\ell} V \xrightarrow{b=E^*K\ell} L^2(X,\mu)$ passes through four distinct spaces. Removing any link: (i) without query, scoring is undefined; (ii) without scoring, no weights, no anchor, no consolidation; (iii) without output, feedback carries no state geometry; (iv) without feedback, M does not influence observation. □

Proposition 14 (Minimal conditions for the consolidation target)**.** *Every theorem in the paper holds under the following conditions on the consolidation target T_j, and no theorem requires more:*

(T1) ***Convexity.*** $T_j \in \mathrm{conv}(M)$ *for all* j.

(T2) ***Anchor positivity.*** $\alpha_v^{(j)} > 0$ *for all* $j \neq v$.

(T3) ***Full positivity.*** $\alpha_k^{(j)} > 0$ *for all* k, j.

(T4) ***Continuity.*** $(M, q) \mapsto T_j(M, q)$ *is continuous.*

(T5) ***Lipschitz.*** $m_j \mapsto T_j$ *is Lipschitz.*

The specific form $T_j = \varphi((m_j + q)M^\top)M$ *satisfies* (T1)–(T5) *but is not uniquely determined by them. It is an instance-defining choice.*

Proof. Each condition is used by at least one theorem.

(T1): Lemma 32 ($\|T_j - v\|^2 \leq \sum_k \alpha_k^{(j)} \|m_k - v\|^2$), Lemma 25 (hull invariance), the capture condition ($q^\top T_j \leq \max_k q^\top m_k$ follows from convexity).

(T2): Lemma 32 (the $k = v$ term contributes zero with positive weight, giving strict contraction), Corollary 33 ($\gamma > 0$ requires $\alpha_v^{(j)} > 0$).

(T3): The consolidation dynamics use strict convexity.

(T4): Corollary 37 (convergence: limit arguments), continuous dependence on initial conditions.

(T5): Local behavior (small changes in m_j produce small changes in T_j).

No proof uses the per-impression dependence on m_j, the additive structure $m_j + q$, the bilinear form $(\cdot)M^\top$, or the reuse of the retrieval φ. Concrete alternatives satisfying (T1)–(T5):

- Global retrieval target: $T_j = \ell = \varphi(qM^\top)M$ (used as simplification in this paper)
- Direct anchor blend: $T_j = (1 - \beta)m_j + \beta v$
- Self-attention without query bias: $T_j = \varphi(m_j M^\top)M$
- Uniform average: $T_j = \frac{1}{|M|} \sum_k m_k$

□

4 Invariance

The components that are not forced turn out to be invariant or boundedly invariant: the loop admits infinite families of equivalent choices, and replacing one with another changes no theorem outside of the simplex map, which is hierarchically defined. This section characterizes the scope of each invariance.

Proposition 15 (Structural invariance; $\mathcal{A}/\mathcal{A}_+/\exp$)**.** *Let $\mathcal{A}$ denote the class of all simplex maps: operators sending bounded measurable $f\colon X \to \mathbb{R}$ to probability measures on X absolutely continuous with respect to μ. Let $\mathcal{A}_+$ denote the subclass of* positive order-preserving *simplex maps (density $a(\tau) > 0$ for μ-a.e. τ whenever f is bounded, and $f(\tau_1) > f(\tau_2) \implies a(\tau_1) > a(\tau_2)$). Let $\mathcal{L}$ denote the class of admissible blend rates: continuous $\lambda\colon [0,1] \to [0,1]$ with $\lambda(0) = 1$, $\lambda(1) = 0$, and $\lambda(r) > 0$ on $[0,1)$.*

(a) ***Forced boundary conditions.*** *The boundary values $\lambda(1) = 0$ and $\lambda(0) = 1$ are forced: violating the first destroys retrieval-gated stability (Lemma 23); violating the second leaves unretrieved impressions permanently frozen.*

(b) ***Invariance.*** *Each result in this paper holds at one of three levels of the containment $\mathcal{A} \supset \mathcal{A}_+ \supset \{\varphi_{\exp}\}$, and states which level it requires. Results with no hypothesis on φ hold for all $(\varphi, \lambda) \in \mathcal{A} \times \mathcal{L}$. Results marked "under any positive φ" hold for all $(\varphi, \lambda) \in \mathcal{A}_+ \times \mathcal{L}$; their proofs use only $w_j > 0$ and not the exponential form. Results marked "under exponential normalization" require the specific form $\varphi(x)_i = \exp(x_i)/\sum_j \exp(x_j)$.*

All results are λ-invariant except where explicitly noted.

Proof. (a) If $\lambda(1) \neq 0$, the anchor $v = \arg\max_k w_k$ is displaced at every consolidation step, and Lemma 23 fails. If $\lambda(0) \neq 1$, an impression with $w_j = 0$ retains a permanent residue of its original position. The qualitative consolidation results depend only on the boundary conditions and $\lambda(r) > 0$ for $r \in [0,1)$, so they hold for any $\lambda \in \mathcal{L}$.

(b) Results at the $\mathcal{A}$ level use only $\int a\, d\mu = 1$ and $a \geq 0$. Results at the $\mathcal{A}_+$ level additionally require $a(\tau) > 0$ μ-a.e. and order-preservation, which gives $w_j > 0$, $0 < \lambda_j < 1$ for non-anchors, and ensures the highest-scored element is the highest-weighted; neither level references the exponential formula. Results at the exponential level use specific properties of exp: the density bound $\exp(-2B)/\mu(X)$ (when $\mu(X) < \infty$), the ratio $\exp(s_i - s_j)$, or the entropy characterization. Exponential normalization additionally requires $\int_X \exp(S')\, d\mu < \infty$; this holds when $\mu(X) < \infty$ or when $S'(\tau) \to -\infty$ sufficiently fast outside a finite-measure set. □

Remark 16. The containment $\mathcal{A} \supset \mathcal{A}_+ \supset \{\varphi_{\exp}\}$ structures the paper: the majority of results hold at the broadest level, a further group requires only positivity, and the quantitative bounds use the exponential form. Each result states which level it requires. The system is also invariant across blend rates: the accumulation scaling $f(t) \sim t$ (Proposition 7) and the capture threshold max (Proposition 10) are forced, and the consolidation results hold for any $\lambda \in \mathcal{L}$. The default $\lambda(r) = 1 - r$ is the simplest admissible interpolant and the unique one for which the displacement-to-deficit ratio $\lambda(r)/(1-r) = 1$ is constant, but no result depends on this specific choice.

Table 1 classifies every non-definition result by its simplex-map requirement. All results are λ-invariant (hold for any admissible blend rate $\lambda \in \mathcal{L}$) and h-invariant (hold for any admissible scoring rule h; Proposition 18).

Table 1: Invariance classification of results. Level $\mathcal{A}$: any simplex map. Level $\mathcal{A}_+$: any positive order-preserving simplex map. Level $\{\exp\}$: the specific exponential form. All results are also λ-invariant and h-invariant.

Result	**Level**	**Key property used**
Rigidity and Invariance (Sections 3–4)		
Prop. 15	$\mathcal{A}_+$	defines the hierarchy
Cor. 19	$\mathcal{A}$	scaling preserves inequality
Prop. 7	$\mathcal{A}$	C/N convergence, no φ needed
Lem. 9	$\mathcal{A}$	set-union argument

Result	Level	Key property used
Prop. 10	$\mathcal{A}$	max monotone under enlargement
Prop. 13	$\mathcal{A}$	data-flow constraints
Prop. 18	$\mathcal{A}$	φ absorbs h-variation
Prop. 11	$\mathcal{A}$; (iv)–(v) $\mathcal{A}_+$/exp	necessity of each step
Prop. 14	$\mathcal{A}$	minimal target conditions
Thm. 12	$\mathcal{A}$	structural classification
Core Dynamics (Section 5)		
Cor. 20	$\mathcal{A}$	deterministic counter
Thm. 21	$\mathcal{A}$	σ recurrence, density-bounded
Lem. 23	$\mathcal{A}_+$	$\lambda_v = 0$, $\lambda_j \in (0,1)$
Cor. 24	$\mathcal{A}$	$\lvert M\rvert$ non-decreasing by union
Lem. 25	$\mathcal{A}$	convex combination
Lem. 26	$\mathcal{A}$	idempotence
Cor. 27	$\mathcal{A}$	$w \in \Delta$, hull bound
Thm. 28	$\mathcal{A}_+$	local positivity; $\exp(-2B)/\mu(X)$ for exp. normalization
Cor. 29	$\mathcal{A}$	$w \in \Delta$
Cor. 30	$\mathcal{A}$	$a \in \Delta$
Cor. 31	$\mathcal{A}_+$	$w_j > 0 \Rightarrow 0 < \lambda_j \leq 1$
Lem. 32	$\mathcal{A}_+$	$\alpha_v > 0$, contraction
Cor. 33	$\mathcal{A}_+$	inherits from Lem. 32
Cor. 34	exp	$\exp(s_{\text{new}})$ in denominator
Cor. 35	exp	$\exp(s_1)/\sum \exp(s_k)$
Cor. 36	$\mathcal{A}$	$\int a\, d\mu = 1$
Cor. 37	$\mathcal{A}_+$	Lyapunov descent; $\lambda(r) \geq c(1-r)$
Cor. 38	$\mathcal{A}_+$	$w_j/w_v \to 1$ under φ_+
Cor. 39	$\mathcal{A}_+$	$V(t) \to 0$
Cor. 40	$\mathcal{A}_+$	contraction rate γ
Carrying Capacity (Section 6)		
Thm. 41	exp	$\lvert M^*\rvert$, $\exp(2R^2)$ bound
Prop. 43	$\mathcal{A}$	state does not feed back into capture control
Cor. 45	exp	fan effect + self-limiting consolidation
Appendix Results		
Prop. 54	$\mathcal{A}$	C^2 lower bound, telescope
Prop. 55	$\mathcal{A}$	$\sigma \to \infty$
Prop. 56	$\mathcal{A}$	$\sigma \to 0$
Support Dynamics (Section 7)		
Thm. 48	$\mathcal{A}$	accumulator freeze, $\sigma \to 0$
Thm. 50	$\mathcal{A}_+$	positivity of φ, $\int a\, d\mu = 1$
Thm. 51	$\mathcal{A}$	$\int a\, d\mu = 1$, σ-finiteness
Thm. 52	$\mathcal{A}$	Thm. 48

Definition 17 (Admissible scoring rule)**.** A scoring rule $h\colon \mathbb{R}^3 \to \mathbb{R}$ is *admissible* if:

(W1) h is C^1 (hence maps bounded inputs to bounded outputs).

(W2) $h(0, \sigma, b)$ depends non-trivially on σ and b (non-degeneracy).

(W3) $h(s, 0, 0)$ is non-degenerate in s (faithful initialization).

Proposition 18 (Invariance of the scoring rule; $\mathcal{A}$).

(a) Every theorem in the paper holds under any admissible $S'(\tau) = h(S(\tau), \sigma(\tau), b(\tau))$, with the same qualitative conclusions.

(b) Under exponential normalization, $h(s, \sigma, b) = s + \sigma + b$ is the unique admissible rule (up to coordinatewise reparametrization) for which the observation ratio factorizes into independent evidence contributions:

$$\frac{a(\tau_1)}{a(\tau_2)} = F_S(S(\tau_1), S(\tau_2)) \cdot F_\sigma(\sigma(\tau_1), \sigma(\tau_2)) \cdot F_b(b(\tau_1), b(\tau_2)).$$

(c) The independent-evidence factorization is a modeling choice, not a consequence of the dynamics.

Proof. (a) Fix any admissible h satisfying (W1)–(W3). We prove every theorem holds.

Step 1: Boundedness of σ and b (independent of h). These quantities do not involve S'. Corollary 27 gives $|b| \leq R^2\|K\|$ via $b = EK(wM)$ and the fact that $w \in \Delta$ and $M \subseteq \mathrm{conv}(E(X))$ (Lemma 25). Since φ is density-bounded and $|b|$ is bounded independently of h, the density a is uniformly bounded by some $A > 0$ once σ is bounded. Theorem 21(iii) then gives $|\sigma| \leq \sigma_{\max}$ via the self-regulating recurrence on C/N.

Step 2: Compactness of the feasible domain. The tuple $(S(\tau), \sigma(\tau), b(\tau))$ satisfies $|S| \leq B_S$ (canonical assumption), $|\sigma| \leq \sigma_{\max}$, $|b| \leq R^2\|K\|$. For each τ, the triple lies in the compact set $\mathcal{K} = [-B_S, B_S] \times [-\sigma_{\max}, \sigma_{\max}] \times [-R^2\|K\|, R^2\|K\|] \subset \mathbb{R}^3$.

Step 3: Continuity and boundedness of a. By (W1), h is C^1, so $S'(\tau) = h(S(\tau), \sigma(\tau), b(\tau))$ is bounded: $|S'(\tau)| \leq B$ for some B depending only on $\mathcal{K}$. For $\varphi \in \mathcal{A}_+$, positivity of the density on bounded inputs gives $a(\tau) > 0$ μ-a.e. On any compact $K \subset X$, the local non-degeneracy (Theorem 28) provides a uniform lower bound $\alpha_K > 0$. For general $\varphi \in \mathcal{A}$, the C^1 condition ensures the density is bounded.

Step 4: Non-degeneracy when $S = 0$. When $S \equiv 0$, condition (W2) guarantees $h(0, \sigma, b)$ depends nontrivially on at least one of σ or b. Since σ varies by Theorem 21 and b varies nontrivially when M contains multiple impressions, a takes multiple distinct values. Thus the blend rates $\lambda_j = \lambda(w_j / \max w)$ are well-defined and drive consolidation.

Step 5: Every theorem requires only these three properties. We verify the requirements:

- **Lower bound on observation weights:** Theorem 28 requires $|a(\tau)| \geq \alpha > 0$. Step 3 provides this for any h.
- **Variation in consolidation:** Lemma 32 and Corollary 38 require w to vary. This follows from a varying (Step 4) and $w = \varphi(qM^\top)$ with $q = aE$.
- **Continuity for convergence:** Theorem 21 and Lemma 32 use continuous function arguments. Since $a = \varphi(h(\cdot))$ is continuous (compositions of C^1 maps), all continuity requirements hold.

All other theorems explicitly quantify over arbitrary probability densities a or use only σ, M, q, w, b, but none require the specific form of S'.

(b) Under exponential normalization, $a(\tau_1)/a(\tau_2) = \exp(h(s_1, \sigma_1, b_1) - h(s_2, \sigma_2, b_2))$. Requiring this to factorize as $F_S \cdot F_\sigma \cdot F_b$ and taking logarithms, the Pexider–Aczél characterization gives $h(s, \sigma, b) = f_S(s) + f_\sigma(\sigma) + f_b(b) + c$. With inputs in log-score coordinates, $f_S = f_\sigma = f_b = \mathrm{id}$ is the canonical choice.

(c) For $\varphi \in \mathcal{A}$ or $\mathcal{A}_+$, no universal multiplicative decomposition of the "observation ratio" exists; the argument in (b) is specific to the exponential family. □

Corollary 19 (Scale invariance of the capture decision; $\mathcal{A}$)**.** *For any $\beta > 0$, the capture decision under query $q' = \beta q$ is identical to that under q: $q'^\top E(\tau) > \max_{k \in M} q'^\top m_k$ if and only if $q^\top E(\tau) > \max_{k \in M} q^\top m_k$.*

Proof. By Definition 6, $E(\tau)$ is captured iff $q^\top E(\tau) > \max_{k \in M} q^\top m_k$. Scaling q by $\beta > 0$: $\beta q^\top E(\tau) > \beta \max_{k \in M} q^\top m_k$ iff $q^\top E(\tau) > \max_{k \in M} q^\top m_k$, since $\beta > 0$ preserves the inequality. □

5 Core Dynamics

We turn to the dynamics. The following results are the building blocks for the later theorems.

Corollary 20 (History-independence of the lifetime counter; $\mathcal{A}$)**.** *For every $\tau \in X$ and every $t \geq 0$, $N_t(\tau) = N_0 + t$. The denominator of $\sigma = C/N$ is independent of the observation history $\{a_s(\tau)\}_{s<t}$.*

Proof. By Definition 3, $N_{t+1}(\tau) = N_t(\tau) + 1$ for all τ, unconditionally. By induction, $N_t(\tau) = N_0 + t$. □

Theorem 21 (Selective persistence; $\mathcal{A}$)**.** *Let $\tau \in X$.*

(i) If $a_t(\tau) = 0$ for all $t \geq T$, then $\sigma_t(\tau) \to 0$.

(ii) If $a_t(\tau) \geq \delta > 0$ for all t, then $\liminf_{t\to\infty} \sigma_t(\tau) \geq \sqrt{\delta}$.

(iii) If there exists $A > 0$ such that $a_t(\tau) \leq A$ for all t (automatic when μ is counting measure on a finite set, with $A = 1$), then $\sigma_t(\tau) = O(1)$.

Proof. (i) $C_t(\tau) = C_T(\tau)$ for $t \geq T$ since the update adds zero. $N_t(\tau) \to \infty$. $\sigma_t = C_T/N_t \to 0$.

(ii) From the recurrence $C_{t+1}^2 = C_t^2 + 2a_t(t+1) + a_t^2(t+1)^2/C_t^2 \geq C_t^2 + 2\delta(t+1)$. Summing: $C_t^2 \geq C_0^2 + \delta\, t(t+1)$. So $C_t \geq \sqrt{\delta\, t(t+1)}$. Since $N_t = N_0 + t$, we have $\sigma_t = C_t/N_t \geq \sqrt{\delta\, t(t+1)}/(N_0 + t) \to \sqrt{\delta}$.

(iii) Since $a \leq A$, $C_{t+1}^2 \leq C_t^2 + 2A(t+1) + A^2(t+1)^2/C_t^2$. From (ii), $C_s = \Omega(s)$ for persistent elements, so the last term is $O(1)$ per step and $C_t = O(t)$. For non-persistent elements, $\sigma \to 0$ by (i). It remains to handle the general case of an arbitrary observation sequence with $a_t \leq A$ (e.g. intermittent observation that is neither eventually zero nor bounded away from zero).

The ratio $\sigma_t = C_t/N_t$ is self-regulating. The increment satisfies $C_{t+1} - C_t = a_t(t+1)/C_t$, so

$$\sigma_{t+1} - \sigma_t = \frac{C_{t+1} - C_t}{N_{t+1}} - \frac{C_t}{N_t\, N_{t+1}} = \frac{1}{N_{t+1}}\left(\frac{a_t(t+1)}{C_t} - \frac{C_t}{N_t}\right) = \frac{1}{N_{t+1}}\left(\frac{a_t(t+1)}{C_t} - \sigma_t\right).$$

When $\sigma_t > K$ for any fixed $K > 0$, we have $C_t = \sigma_t N_t > K N_t$, so $a_t(t+1)/C_t \leq A(t+1)/(K N_t)$. Since $N_t = N_0 + t$, for t large enough $(t+1)/N_t \leq 2$, giving $a_t(t+1)/C_t \leq 2A/K$. Meanwhile $\sigma_t > K$. Choosing $K > \sqrt{2A}$ ensures the parenthesized term is negative for all large t, so $\sigma_{t+1} < \sigma_t$. Therefore σ_t cannot grow without bound, and $\sigma_t = O(1)$. □

Remark 22 (Density bound). When μ is counting measure on a finite set, $a(\tau) \leq 1$ for all τ and t, so the hypothesis of (iii) holds with $A = 1$. For general (X, μ) with φ density-bounded, the bound $A = A(\|S'\|_\infty)$ depends on $\sigma_{\max}$, which is what (iii) establishes. The circularity resolves at each finite step: $\sigma_0 = C_0/N_0$ is given finite, so a_0 is bounded, so σ_1 is finite, and by induction $\sigma_t < \infty$ for all t. The self-regulating argument then shows $\sup_t \sigma_t < \infty$ whenever a uniform density bound exists. For the remainder of the paper, we write $\sigma_{\max}$ for this bound, noting that it is automatic in the finite case and is a well-posedness condition in the general case.

Lemma 23 (Retrieval-gated stability; $\mathcal{A}_+$). *Let $v = \arg\max_{k \in M} w_k$ at step t. Then v is unchanged by consolidation, and the capture threshold in v's direction never decreases.*

Proof. $\lambda_v = 0$, so consolidation preserves v exactly. For any q' with $v = \arg\max_{k \in M_t} q'^\top m_k$,

$$\max_{k \in M_{t+1}} q'^\top m_k \;\geq\; \max_{k \in M_t} q'^\top m_k,$$

since $v \in M_{t+1}$ unchanged and $M_{t+1} \supseteq \{v\}$. □

Corollary 24 (Monotone acquisition; $\mathcal{A}$). *$|M|$ is non-decreasing: capture adds elements by union (Definition 6) and no operation removes them. The capture threshold $\max_{k \in M} q^\top m_k$ is non-decreasing in $|M|$.*

Proof. Capture adds elements to M by union, so $|M|$ is non-decreasing. The threshold $\max_{k \in M} q^\top m_k$ is non-decreasing in $|M|$ since adding elements can only increase or maintain the maximum. □

Lemma 25 (Convex hull invariance; $\mathcal{A}$). *For all t, $\mathrm{conv}(M_t) \subseteq \overline{\mathrm{conv}}(E(X))$. In particular, $\|m_j\| \leq R$ for every row of M_t.*

Proof. At $t = 0$, $M_0 = \emptyset$ and the inclusion holds vacuously. Capture admits only vectors of the form $E(\tau)$, so after every capture step $M \subseteq E(X) \subseteq \overline{\mathrm{conv}}(E(X))$. For consolidation, each row of the target $T_j = \sum_i [\varphi((M+q)\, M^\top)]_{ji}\, m_i$ is a convex combination of rows of M, so $T_j \in \mathrm{conv}(M)$. Each updated row $M'_j = (1 - \lambda_j)\, m_j + \lambda_j\, T_j$ is a convex combination of two points in $\mathrm{conv}(M)$, so $M'_j \in \mathrm{conv}(M)$. By induction, $\mathrm{conv}(M_t) \subseteq \overline{\mathrm{conv}}(E(X))$ for all t. Since $\|E(\tau)\| \leq R$ for all τ, every point in $\overline{\mathrm{conv}}(E(X))$ has norm at most R. □

Lemma 26 (Bounded state without consolidation; $\mathcal{A}$). *If consolidation is disabled (i.e. $\lambda \equiv 0$), then (i) each vector $E(\tau)$ enters M at most once, (ii) $M_t \subseteq M_0 \cup E(X)$, so $|M_t| \leq |M_0| + |X|$ when X is finite, and (iii) M_t is finite for all t provided the capture set at each step is finite (automatic when X is finite).*

Proof. Without consolidation, members of M are never modified, so M contains only vectors of the form $E(\tau)$ for $\tau \in X$. If $E(\tau) \in M_t$ already, then $\max_{k \in M_t} q^\top m_k \geq q^\top E(\tau)$, so the strict inequality in Definition 6 fails and $E(\tau)$ is not re-admitted. This proves (i). Part (ii) follows: every element of M_t belongs to $M_0 \cup E(X)$, and no embedding is counted twice. Part (iii): for finite X, $|M_t| \leq |M_0| + |X| < \infty$. For infinite X, finiteness holds provided each capture step $\{E(\tau) : q^\top E(\tau) > \theta_t\}$ is a finite set, which is automatic when the level sets of $\tau \mapsto q^\top E(\tau)$ have finite cardinality for each q. □

Corollary 27 (Bounded retrieval bias; $\mathcal{A}$). *$|b(\tau)| \leq R^2 \|K\|_{\mathrm{op}}$ for all τ and all t.*

Proof. By Lemma 25, every row of M lies in $\mathrm{conv}(E(X))$, so $\|m_j\| \leq R$ for all j. Since $\ell = wM$ with w a convex weight vector, $\|\ell\| \leq R$. Then $|b(\tau)| = |E(\tau)^\top (K\ell)| \leq \|E(\tau)\| \cdot \|K\|_{\mathrm{op}} \cdot \|\ell\| \leq R^2 \|K\|_{\mathrm{op}}$. □

Theorem 28 (Non-degeneracy; $\mathcal{A}_+$). *Let $\varphi \in \mathcal{A}_+$ and suppose $|S(\tau)| \leq B_S$ for all $\tau \in X$ and all t.*

(i) ***Local non-degeneracy.*** *For any measurable $K \subseteq X$ with $0 < \mu(K) < \infty$, there exists $\alpha_K > 0$ such that $a_t(\tau) \geq \alpha_K$ for μ-a.e. $\tau \in K$ and all $t \geq 0$. For exponential normalization with $\mu(X) < \infty$: $\alpha_K \geq \exp(-2B)/\mu(X)$ where $B = B_S + \sigma_{\max} + R^2\|K\|_{\mathrm{op}}$. For exponential normalization with $\mu(X) = \infty$ (requiring $Z_t < \infty$): $\alpha_K \geq \exp(-B)/\sup_t Z_t$.*

(ii) ***Global degeneracy.*** *If $\mu(X) = \infty$, then for every $\varepsilon > 0$ and every t, the set $\{\tau : a_t(\tau) \geq \varepsilon\}$ has finite measure: $\mu(\{\tau : a_t(\tau) \geq \varepsilon\}) \leq 1/\varepsilon$. In particular, $a_t(\tau) \to 0$ along any sequence τ_n with $\mu(B(\tau_n, r)) \to \infty$ for fixed r.*

(iii) ***Essential support concentration.*** *For any $\varepsilon > 0$ and any t, there exists a set $K_\varepsilon \subseteq X$ with $\mu(K_\varepsilon) < \infty$ such that $\int_{X \setminus K_\varepsilon} a_t \, d\mu < \varepsilon$. When $\mu(X) < \infty$ and φ is exponential normalization, K_ε can be chosen independently of t (the family $\{a_t \, d\mu\}$ is uniformly tight).*

Proof. By Theorem 21(iii), there exists $\sigma_{\max}$ such that $|\sigma_t(\tau)| \leq \sigma_{\max}$ for all τ, t. By Corollary 27, $|b_t(\tau)| \leq R^2\|K\|_{\mathrm{op}}$ for all τ, t. Set $B = B_S + \sigma_{\max} + R^2\|K\|_{\mathrm{op}}$. Then $|S'_t(\tau)| \leq B$ for all τ, t.

(i) Since $\varphi \in \mathcal{A}_+$ is positive on bounded inputs and $|S'(\tau)| \leq B$, the density satisfies $a_t(\tau) > 0$ for μ-a.e. τ. For the quantitative bound under exponential normalization (which requires $Z_t := \int_X \exp(S'_t) \, d\mu < \infty$; this holds when $\mu(X) < \infty$ or when $S'(\tau) \to -\infty$ sufficiently fast): on any K with $0 < \mu(K) < \infty$,

$$a_t(\tau) = \frac{\exp(S'_t(\tau))}{Z_t} \geq \frac{\exp(-B)}{Z_t}.$$

When $\mu(X) < \infty$: $Z_t \leq \int_X \exp(B) \, d\mu = \exp(B) \, \mu(X)$, so $a_t(\tau) \geq \exp(-2B)/\mu(X)$ on K. When $\mu(X) = \infty$: $Z_t < \infty$ by assumption, so $a_t(\tau) \geq \exp(-B)/Z_t > 0$; the bound depends on Z_t. For general $\varphi \in \mathcal{A}_+$: on any fixed K with $\mu(K) < \infty$, the density is bounded below by a positive constant α_K depending on K, B, and φ.

(ii) Since $\int_X a_t \, d\mu = 1$, Markov's inequality gives $\mu(\{a_t \geq \varepsilon\}) \leq 1/\varepsilon$.

(iii) For general $\varphi \in \mathcal{A}_+$: fix t and let K_n be an exhaustion of X by finite-measure sets ($K_1 \subset K_2 \subset \cdots$, $\bigcup K_n = X$). Then $\int_{K_n} a_t \, d\mu \nearrow 1$ by monotone convergence. Choose n large enough that $\int_{X \setminus K_n} a_t \, d\mu < \varepsilon$; this n may depend on t. For exponential normalization with $\mu(X) < \infty$: the upper bound $a_t(\tau) \leq \exp(2B)/\mu(X)$ (since $Z_t \geq \exp(-B) \, \mu(X)$) is t-independent, so $\int_{X \setminus K_n} a_t \, d\mu \leq \exp(2B) \, \mu(X \setminus K_n)/\mu(X)$ uniformly in t. Since $\mu(X \setminus K_n) \to 0$, choosing n with $\mu(X \setminus K_n) < \varepsilon \, \mu(X) \exp(-2B)$ gives a t-independent K_ε. □

Corollary 29 (Retrieval output lies in convex hull; $\mathcal{A}$)**.** *For any query q with $M \neq \emptyset$, the retrieval output $\ell = wM$ satisfies $\ell \in \mathrm{conv}(M) \subseteq \mathrm{conv}(E(X))$.*

Proof. By Definition 4, $w = \varphi(qM^\top) \in \Delta^{|M|}$, so $\ell = \sum_j w_j m_j$ is a convex combination of rows of M. Hence $\ell \in \mathrm{conv}(M)$. By Lemma 25, $\mathrm{conv}(M) \subseteq \mathrm{conv}(E(X))$. □

Corollary 30 (Context vector norm bound; $\mathcal{A}$)**.** *For any step t, the context vector satisfies $\|q_t\| \leq R$. Moreover, $q_t \in \overline{\mathrm{conv}}(E(X))$ (the closed convex hull of the embedding image).*

Proof. By Definition 4, $q = \int_X a(\tau) \, E(\tau) \, d\mu(\tau)$ with $\int a \, d\mu = 1$ and $a \geq 0$. By convexity of the norm and Jensen's inequality, $\|q\| \leq \int a(\tau)\|E(\tau)\| \, d\mu(\tau) \leq R \int a \, d\mu = R$. The integral is a limit of convex combinations, so $q \in \overline{\mathrm{conv}}(E(X))$. □

Corollary 31 (Bounds on the consolidation blend rate; $\mathcal{A}_+$)**.** *Under any positive φ with $|M| \geq 2$ and $v = \arg\max_k w_k$ the unique maximizer of the retrieval weights, every blend rate satisfies $0 \leq \lambda_j \leq 1$, with $\lambda_v = 0$ for the anchor and $0 < \lambda_j \leq 1$ for all $j \neq v$.*

Proof. By Definition 5, $\lambda_j = \lambda(w_j / \max_k w_k)$. Since $\lambda \colon [0,1] \to [0,1]$ with $\lambda(1) = 0$ and $\lambda(0) = 1$, we have $0 \leq \lambda_j \leq 1$. For the anchor $v = \arg\max_k w_k$: $\lambda_v = \lambda(1) = 0$. For $j \neq v$: since φ is positive, $w_j > 0$, so $w_j/w_v > 0$ and $\lambda_j \leq \lambda(0) = 1$. Since v is the unique maximizer, $w_j < w_v$, so $w_j/w_v < 1$ and $\lambda_j > \lambda(1) = 0$. □

Lemma 32 (Lyapunov function for fixed-context consolidation; $\mathcal{A}_+$)**.** *Let $\varphi \in \mathcal{A}_+$. Fix a query q and suppose capture is disabled. Let $v = \arg\max_{k \in M} w_k$ where $w = \varphi(qM^\top)$, and $V(t) = \max_{j \neq v} \|m_j - v\|$. Then $V(t+1) < V(t)$ whenever $V(t) > 0$.*

Proof. Since $\varphi \in \mathcal{A}_+$ is order-preserving, $v = \arg\max_k w_k = \arg\max_k q^\top m_k$. Since $\lambda_v = 0$, the anchor v is unchanged. For $j \neq v$, the consolidation target is $T_j = \sum_k \alpha_k m_k$ where $\alpha = \varphi((m_j + q)\, M^\top)$ has all entries positive (since φ is positive). Then $T_j - v = \sum_k \alpha_k (m_k - v)$, and since $\alpha_v > 0$ contributes the zero vector:

$$\|T_j - v\|^2 = \Big\|\sum_k \alpha_k (m_k - v)\Big\|^2 \leq \sum_k \alpha_k \|m_k - v\|^2 \leq V(t)^2$$

with strict inequality (the $k = v$ term contributes zero with positive weight). Since $\lambda_j > 0$ for $j \neq v$:

$$\|M'_j - v\| \leq (1 - \lambda_j)\|m_j - v\| + \lambda_j \|T_j - v\| < (1 - \lambda_j)\, V(t) + \lambda_j\, V(t) = V(t).$$

Taking the maximum over $j \neq v$ gives $V(t+1) < V(t)$. □

Corollary 33 (State-dependent contraction rate; $\mathcal{A}_+$)**.** *Under the hypotheses of Lemma 32, let $\alpha_v^{(j)}$ denote the weight that member j's consolidation target places on the anchor v. Then*

$$V(t+1) \leq (1 - \gamma(t))\, V(t), \qquad \gamma(t) = \min_{j \neq v} \lambda_j \big(1 - \sqrt{1 - \alpha_v^{(j)}}\big).$$

Proof. From the proof of Lemma 32, $\|T_j - v\|^2 \leq \sum_k \alpha_k^{(j)} \|m_k - v\|^2 \leq (1 - \alpha_v^{(j)})\, V(t)^2$, so $\|T_j - v\| \leq \sqrt{1 - \alpha_v^{(j)}}\, V(t)$. Then $\|M'_j - v\| \leq (1 - \lambda_j)\, V(t) + \lambda_j \sqrt{1 - \alpha_v^{(j)}}\, V(t) = (1 - \lambda_j(1 - \sqrt{1 - \alpha_v^{(j)}}))\, V(t)$. □

Corollary 34 (Retrieval weight dilution (fan effect); exp. normalization)**.** *Let $\varphi = \varphi_{\exp}$ and q be fixed. If $M' = M \cup \{m_{\text{new}}\}$ for some $m_{\text{new}} \notin M$, then for every $j \in M$:*

$$[\varphi_{\exp}(q\, M'^\top)]_j < [\varphi_{\exp}(q\, M^\top)]_j.$$

In particular, $\max_{j \in M} w_j$ is strictly decreasing as $|M|$ grows.

Proof. Let $s_j = q^\top m_j$. For $j \in M$:

$$[\varphi_{\exp}(qM^\top)]_j = \frac{\exp(s_j)}{\sum_{k \in M} \exp(s_k)}.$$

Adding m_{new} with score $s_{\text{new}} = q^\top m_{\text{new}}$ increases the denominator by $\exp(s_{\text{new}}) > 0$ while the numerator is unchanged. Hence every existing weight strictly decreases. □

Corollary 35 (Retrieval weight bounds; exp. normalization)**.** *Under exponential normalization with $|M| = n$ and score gap $g = q^\top m_v - \max_{k \neq v} q^\top m_k > 0$, the leading retrieval weight satisfies*

$$w_v \;\geq\; \frac{1}{1 + (n-1)\exp(-g)}.$$

This worst case is attained when all non-leading scores are equal. The bound requires $g = \Omega(\log n)$ for w_v to remain bounded away from zero: maintaining $w_v \geq c$ for a fixed $c > 0$ requires $g \geq \log((n-1)c/(1-c))$.

Proof. Write $w_v = 1/(1+\sum_{k\neq v}\exp(s_k - s_v))$. Since $s_k \leq s_v - g$ for all $k \neq v$, each $\exp(s_k - s_v) \leq \exp(-g)$, and $\sum_{k\neq v}\exp(s_k - s_v) \leq (n-1)\exp(-g)$. Therefore $w_v \geq 1/(1+(n-1)\exp(-g))$. Setting this $\geq c$ and solving: $(n-1)\exp(-g) \leq (1-c)/c$, so $g \geq \log((n-1)c/(1-c)) = \Omega(\log n)$. □

Corollary 36 (Observation budget constraint; $\mathcal{A}/\mathcal{A}_+$)**.** *Under any simplex map φ, $\int_X a_t\, d\mu = 1$ at every step. For any measurable $A \subseteq X$, the total attention allocated to A is $\int_A a_t\, d\mu \leq 1$. Under $\mathcal{A}_+$, enlarging the domain (replacing X by $X' \supset X$ with $\mu(X' \backslash X) > 0$) strictly dilutes the density on any fixed subset of positive measure.*

Proof. φ maps to a probability measure, so $\int_X a_t\, d\mu = 1$ by definition. The constraint $\int_A a_t\, d\mu \leq 1$ is immediate. If $X' = X \cup Y$ with $\mu(Y) > 0$, then $\int_{X'} a'_t\, d\mu = 1$ and $\int_Y a'_t\, d\mu > 0$ (under $\mathcal{A}_+$), so $\int_X a'_t\, d\mu < 1 = \int_X a_t\, d\mu$. □

Corollary 37 (Convergence under repeated consolidation without capture; $\mathcal{A}_+$)**.** *Let φ be a continuous positive simplex map and suppose $\lambda(r) \geq c(1-r)$ near $r = 1$ for some $c > 0$ (satisfied by $\lambda(r) = 1 - r$ and all polynomial blend rates). Suppose capture is disabled, the query q is fixed, and $v = \arg\max_k q^\top m_k$ is unique. Then $V(t) \to 0$: all impressions converge to the anchor v.*

Proof. By Lemma 32, $V(t+1) < V(t)$ whenever $V(t) > 0$. Since $V(t)$ is non-negative, it converges to some $L \geq 0$. Suppose $L > 0$. The trajectory of M is confined to the compact set $\overline{\mathrm{conv}}(E(X)) \cap \overline{B}(0, R)$ (Lemma 25; compact since it is closed and bounded in $\mathbb{R}^d$), so M has a convergent subsequence; since $V(t)$ is strictly decreasing and bounded, M in fact converges to a limit configuration M^* with all items at distance exactly L or less from v.

At M^*, the query q is fixed, so the scores $q^\top m_j^*$ are well-defined. We claim $q^\top v > q^\top m_j^*$ for all $j \neq v$ with $m_j^* \neq v$. Suppose for contradiction that $q^\top m_j^* = q^\top v$ for some $j \neq v$. If there exists any k with $q^\top m_k^* < q^\top v$, then the consolidation target $T_j^* = \sum_k \alpha_k^{(j)} m_k^*$ satisfies $q^\top T_j^* < q^\top v$ (since $\alpha_k^{(j)} > 0$ by positivity), so after one consolidation step $q^\top m'_j < q^\top v$, contradicting M^* being a limit configuration. If instead $q^\top m_k^* = q^\top v$ for *all* k, then all retrieval weights equalize ($w_k = 1/|M|$), so $\lambda_j = \lambda(1) = 0$ and M^* is a fixed point of the dynamics. But V is strictly decreasing along the trajectory ($\gamma(t) > 0$ for all finite t, since the initial unique-maximizer gap ensures $w_j(t) < w_v(t)$ at each finite step); for $V(t) \to L > 0$ at such a fixed point we would need $\sum_t \gamma(t) < \infty$. Under the standard condition $\lambda(r) \geq c(1-r)$ near $r = 1$ (satisfied by $\lambda(r) = 1-r$ and all polynomial blend rates), one shows $\gamma(t) \geq c'/t$ and $\sum \gamma(t) = \infty$, ruling out this case.

Therefore at the limit, $w_j < w_v$ for all $j \neq v$, giving $\lambda_j > 0$. Since $w = \varphi(q\, M^\top)$ depends continuously on M and the set of configurations near M^* is compact (contained in $\overline{\mathrm{conv}}(E(X))$), the strict inequalities $w_j < w_v$ yield a uniform gap: there exists $\delta > 0$ such that $w_j(t) \leq (1-\delta)\, w_v(t)$ for all sufficiently large t. Combined with $\alpha_v^{(j)} > 0$ (positivity of φ), Corollary 33 gives $\gamma_L > 0$, so $V(t+1) \leq (1-\gamma_L)\, V(t)$ for sufficiently large t. This forces $V(t) \to 0$, contradicting $L > 0$. Hence $L = 0$. □

Corollary 38 (Vanishing contraction rate; $\mathcal{A}_+$)**.** *Under the hypotheses of Corollary 37 (fixed query, no capture), the contraction rate $\gamma(t)$ satisfies $\gamma(t) \to 0$ as $t \to \infty$.*

Proof. By Corollary 37, $V(t) \to 0$, so $m_j \to v$ for all j. Since $w_j = [\varphi(q\, M^\top)]_j$ and $q^\top m_j \to q^\top v$ for all j, the retrieval weights equalize (by continuity of φ): $w_j/w_v \to 1$, hence $\lambda_j = \lambda(w_j/w_v) \to \lambda(1) = 0$ by continuity of λ. The factor $1 - \sqrt{1 - \alpha_v^{(j)}}$ remains bounded in $[0, 1]$, so $\gamma(t) = \min_{j\neq v} \lambda_j (1 - \sqrt{1 - \alpha_v^{(j)}}) \to 0$. □

Corollary 39 (Prototype formation; $\mathcal{A}_+$)**.** *Under the hypotheses of Corollary 37, as $t \to \infty$ every row of M converges to the anchor $v = \arg\max_{k\in M} q^\top m_k$.*

Proof. By Corollary 37, $V(t) = \max_{j \neq v} \|m_j - v\| \to 0$, so $m_j \to v$ for all j. □

Corollary 40 (Context variation prevents convergence to single prototype; $\mathcal{A}_+$)**.** *Let φ be a continuous positive simplex map and $|M| \geq 2$ with at least two distinct impressions. Under fixed query q, $V(t) \to 0$ (Corollary 37). Let T_{rot} be the timescale on which the query varies (i.e. the anchor identity changes), and let $\gamma(t) = \min_{j \neq v} \lambda_j (1 - \sqrt{1 - \alpha_v^{(j)}})$ be the contraction rate (Corollary 33). Suppose $\gamma_{\min} := \min_{0 \leq t \leq T_{\text{rot}}} \gamma(t) > 0$ (which holds whenever distinct impressions have distinct scores under the query throughout $[0, T_{\text{rot}}]$). If $T_{\text{rot}} < 1/\gamma_{\min}$, the anchor changes before V reaches zero, and distinct impressions are preserved.*

Proof. By Corollary 37, fixed-query consolidation drives $V \to 0$. By Corollary 33, $V(t+1) \leq (1 - \gamma(t))V(t)$, so after k steps under a fixed anchor, $V \leq (1 - \gamma_{\min})^k V(0)$. Under a fixed anchor, V decreases monotonically. When the anchor changes, however, V is redefined with respect to the new anchor v': $V' = \max_{j \neq v'} \|m_j - v'\|$. If $v' \neq v$, then V' may exceed V because the old anchor v (which had $\|m_j - v\|$ small for all j) is now measured from a different reference point. Specifically, $\|v - v'\| > 0$ whenever $v \neq v'$, so $V' \geq \|v - v'\| > 0$ even if the pre-rotation V was small. When $T_{\text{rot}} < 1/\gamma_{\min}$, each anchor period reduces V by a bounded factor, and the subsequent anchor change resets V' to at least $\|v - v'\|$. This prevents V from converging to zero across rotations, provided the anchor set contains at least two distinct points that are not driven together. □

6 Carrying Capacity

Theorem 41 (Retrieval degradation and carrying capacity; exp. normalization)**.** *Let $\varphi = \varphi_{\text{exp}}$ and let the system operate with both capture and consolidation enabled. Recall $R = \sup_{\tau \in X} \|E(\tau)\| < \infty$ and suppose $\|K\|_{\text{op}} \leq 1$. Then:*

(i) Every capture event strictly decreases all existing retrieval weights: if $M' = M \cup \{m_{\text{new}}\}$, then $[\varphi_{\text{exp}}(qM'^\top)]_j < [\varphi_{\text{exp}}(qM^\top)]_j$ for all $j \in M$.

(ii) The score gap required for a single impression to achieve retrieval weight $w_v \geq c > 0$ is $g \geq \log((|M| - 1)c/(1 - c))$, i.e. $g = \Omega(\log |M|)$.

(iii) The achievable score gap is uniformly bounded: $g \leq 2R^2$.

(iv) There exists a critical state size

$$|M^*| = 1 + \frac{1 - c}{c} \exp(2R^2)$$

beyond which no single impression can achieve retrieval weight $w_v \geq c$ for any query.

(v) Consolidation cannot prevent this degradation: as impressions equalize under consolidation, the retrieval weights converge ($w_j/w_v \to 1$), so $\lambda_j \to 0$ and $\gamma(t) \to 0$. Consolidation decelerates precisely when it is most needed.

Proof. (i) This is exactly Corollary 34. Adding m_{new} to M increases the normalizing denominator by $\exp(q^\top m_{\text{new}}) > 0$ while leaving each numerator $\exp(q^\top m_j)$ unchanged, so every existing weight strictly decreases.

(ii) By Corollary 35, the worst-case retrieval weight under exponential normalization with $|M| = n$ and score gap g satisfies

$$w_v \;\geq\; \frac{1}{1 + (n - 1)\exp(-g)}.$$

For $w_v \geq c$ to be guaranteed (even in the worst case), we need $(n-1)\exp(-g) \leq (1-c)/c$, so $g \geq \log((n-1)c/(1-c)) = \Omega(\log n)$.

(iii) The score gap is $g = q^\top m_v - \max_{k \neq v} q^\top m_k$. By Corollary 30, $\|q\| \leq R$ since $q = aE$ is a convex combination of embeddings. By Lemma 25, every $m_j \in \mathrm{conv}(E(X))$, so $\|m_j\| \leq R$. Therefore $|q^\top m_j| \leq \|q\| \cdot \|m_j\| \leq R^2$ for all j. The maximum possible score gap is thus $g \leq R^2 - (-R^2) = 2R^2$.

(iv) Combining (ii) and (iii): for $w_v \geq c$ we need $g \geq \log((|M|-1)c/(1-c))$, but $g \leq 2R^2$. The condition fails when $\log((|M|-1)c/(1-c)) > 2R^2$, i.e. when

$$|M| > 1 + \frac{1-c}{c}\exp(2R^2) =: |M^*|.$$

For $|M| > |M^*|$, no query can achieve $w_v \geq c$ for any single impression. The bound $|M^*|$ is finite and depends only on c and R. Since $|M|$ is non-decreasing during capture (Corollary 24), the system inevitably crosses $|M^*|$ if it continues to operate without consolidation.

(v) Could consolidation prevent this by reducing the effective size of M? During operation with query variation, consolidation operates under a query q that shifts at every step. The contraction rate $\gamma(t)$ from Corollary 33 satisfies $\gamma(t) = \min_{j \neq v} \lambda_j(1 - \sqrt{1 - \alpha_v^{(j)}})$. As impressions converge under consolidation, $w_j/w_v \to 1$, so $\lambda_j = \lambda(w_j/w_v) \to \lambda(1) = 0$ and $\gamma(t) \to 0$. Consolidation is self-limiting: it decelerates as impressions become similar. Moreover, during operation the query varies, and by Corollary 40, if the anchor rotates faster than $1/\gamma$, distinct impressions are preserved. Consolidation therefore cannot permanently reduce the effective state size below $|M^*|$: it neither removes rows from M nor merges them exactly (it moves them closer but $\gamma \to 0$ before convergence completes, and anchor rotation protects different members at different times). □

Remark 42 (Carrying capacity as a structural constraint). The degradation in Theorem 41 is caused by the interaction of three mechanisms: capture grows $|M|$ monotonically, the fan effect (Corollary 34) dilutes all weights at each capture event, and the bounded score gap prevents the system from overcoming the dilution. The bound $|M^*|$ is the carrying capacity of the retrieval mechanism under a given embedding. For unit-norm embeddings ($R = 1$), $|M^*| = 1 + ((1-c)/c)e^2$. At $c = 0.5$ (the majority retrieval threshold, where the anchor outweighs all competitors combined) this gives $|M^*| = 1 + e^2 \approx 8.4$.

Importantly, the carrying capacity theorem is φ-invariant and λ-invariant: it holds for any simplex map and any blend rate. Yet the loop cannot implement the internal switch needed to respect this capacity. This reveals a deeper connection: the invariance of the dynamics and the necessity of external control are two aspects of the same structural revelation. The loop's behavior is forced by the architecture, but its self-preservation is not forced. The loop knows when to stop growing, but it lacks the internal machinery to act on that knowledge.

Proposition 43 (Necessity of an external capture disable; $\mathcal{A}$)**.** *The loop cannot autonomously disable capture. The capture-enable condition (step 5 of the loop in Definition 6) depends on the external score $S(\tau)$, which is an input to the system, not a state variable: no step of the five-step loop modifies S or the capture threshold θ. Therefore, even when the carrying capacity condition is satisfied ($|M| \geq |M^*|$), the system continues capturing indefinitely unless an external controller monitors $|M|$ (or any proxy such as the retrieval concentration $\rho_t = \max_j w_j$) and disables capture.*

Proof. By inspection of the five-step loop (Definitions 1–6): the system's state is (C, N, M, ℓ); its inputs are S and any external capture control flag. Step 4 (consolidate) updates only M. Step 5 (capture) adds rows to M when $q^\top E(\tau) > \theta$, where $\theta = \max(qM^\top)$. Neither step modifies S, the capture threshold computation, or any external control flag. The observable quantities $(|M|, \rho_t)$ are computed at each step but do not feed back into S or the capture-enable condition.

Therefore, a system operating with capture enabled remains in that configuration at the next step, regardless of whether $|M| \geq |M^*|$. □

Remark 44 (Mathematical closure vs. control incompleteness). The loop is operationally closed (five steps, all necessary) but control-theoretically open (cannot autonomously disable capture). This reveals a conceptual distinction: a system can be *mathematically* complete while remaining *behaviorally* incomplete. The signals needed for capture control exist internally (the condition $|M| \geq |M^*|$ can be computed at every step), but the causal pathways to act on them do not. An external controller supplies the missing bridge. This suggests a structural principle: signal generation and control execution belong to different conceptual layers in self-referential dynamical systems. The loop can compute the capacity condition internally but lacks pathways to act on it.

Corollary 45 (Cost of ignoring the capacity limit; exp. normalization)**.** *Suppose the capacity condition $|M| \geq |M^*|$ is satisfied at time t_0 but capture remains enabled for k additional steps. Then:*

(i) $|M|$ continues to grow monotonically (Corollary 24), with ρ_t degrading further by the fan effect (Corollary 34).

(ii) No waking consolidation can reverse this degradation: as impressions equalize under the capture pressure, $w_j/w_v \to 1$ and $\lambda_j \to 0$, so $\gamma_t \to 0$. The contraction rate vanishes precisely when consolidation would be most needed.

(iii) If capture is ignored indefinitely ($k \to \infty$), the system enters permanent degradation: $\rho_t \to 0$ as $|M| \to \infty$ with no bound on achievable retrieval weight.

Proof. (i) Corollary 24 shows $|M|$ is non-decreasing; Corollary 34 shows each capture event strictly dilutes all existing retrieval weights. *(ii)* Corollary 33 gives the contraction rate $\gamma(t) = \min_{j \neq v} \lambda_j (1 - \sqrt{1 - \alpha_v^{(j)}})$. As $|M|$ grows, the fan effect (Corollary 34) flattens the retrieval weights: $w_j/w_v \to 1$, so $\lambda_j = \lambda(w_j/w_v) \to 0$ and $\gamma(t) \to 0$. Consolidation cannot accelerate to counteract the capture growth. *(iii)* As $|M| \to \infty$, Theorem 41(ii) gives the score gap requirement $g \geq \Omega(\log |M|)$, but the achievable gap is bounded by $2R^2$ (part (iii)). The condition for $w_v \geq c$ fails for any fixed c once $|M| > |M^*|$, and $\rho_t \to 0$. With $\gamma_t \to 0$, consolidation cannot prevent this. The system is permanently stuck. □

The external controller is not an additional assumption. The loop computes the condition $|M| \geq |M^*|$ at every step, so the signal for self-regulation exists internally. What is missing is the causal pathway to act on it. The loop is mathematically closed (all five feedback operations are explicit) yet control-theoretically open (it cannot autonomously switch regimes). This suggests a more general principle. A self-referential system can be complete at the level of dynamics but incomplete at the level of self-preservation. Signal generation and control execution belong to different structural layers. The architecture forces the behavior, but it does not force self-preservation. This structural separation may be a fundamental property of self-referential dynamical systems rather than a peculiarity of this specific object.

Remark 46 (Regime where structural tension is exposed). The loop faces a structural asymmetry: it can observe the entire domain X at every step, but it can only store at most $|M^*|$ impressions. The observation capacity is unbounded, the storage capacity is bounded. This asymmetry is forced by the loop's architecture. When $|X| \leq |M^*|$, the storage bound is trivially satisfied, and the selection mechanism never genuinely operates. When $|X| \gg |M^*|$, the loop must choose what to save at every step. The infinite regime X with $\mu(X) = \infty$ is the setting where this structural tension is fully exposed. All results in this paper hold on any σ-finite measure space, but the nontrivial interplay between unbounded observation and bounded storage only manifests when

the observation space exceeds the carrying capacity. We suspect a deeper structural principle may explain why the loop emerges naturally in settings where $|X| \gg |M^*|$, but we leave this investigation to future work.

7 Support Dynamics

When X is finite, the active set $\{\tau : a_t(\tau) > 0\}$ is always X under any positive simplex map. When X is an infinite measure space, the budget constraint $\int a\, d\mu = 1$ forces concentration, and the distinction between $\mathcal{A}$ and $\mathcal{A}_+$ becomes qualitatively sharper. The phenomena in this section do not exist in the finite case.

Definition 47 (Active set)**.** The *active set* at time t is $A_t = \{\tau \in X : a_t(\tau) > 0\}$, defined up to μ-null sets.

Theorem 48 (Consequences of zero observation; $\mathcal{A}$)**.** *Let $\varphi \in \mathcal{A}$ (not necessarily positive). Suppose $a_s(\tau_0) = 0$ for all $s \geq t$ for some t and $\tau_0 \in X$. Then:*

(i) ***Accumulator freeze.*** *$C_s(\tau_0) = C_t(\tau_0)$ for all $s \geq t$.*

(ii) ***Salience decay.*** *$\sigma_s(\tau_0) \to 0$ as $s \to \infty$.*

(iii) ***Decoupling.*** *The query q_s is independent of $E(\tau_0)$ for all $s \geq t$.*

Proof. (i) Since $a_s(\tau_0) = 0$ for $s \geq t$, the accumulator update adds 0 at each step: $C_{s+1}(\tau_0) = C_s(\tau_0)$ for all $s \geq t$.

(ii) $N_s(\tau_0) = N_t(\tau_0) + (s - t) \to \infty$, while $C_s(\tau_0) = C_t(\tau_0)$ is constant by (i). Hence $\sigma_s(\tau_0) = C_t(\tau_0)/N_s(\tau_0) \to 0$.

(iii) The query $q_s = \int a_s\, E\, d\mu$. Since $a_s(\tau_0) = 0$, the integrand vanishes at τ_0, so q_s is independent of $E(\tau_0)$. All downstream quantities (w_s, ℓ_s, M_s) depend on q_s and not directly on τ_0. □

Remark 49 (Irreversibility of forgetting). Parts (i)–(iii) are unconditional: they hold for any $\varphi \in \mathcal{A}$. Whether $a_s(\tau_0)$ *remains* zero for all $s > t$ depends on φ. For threshold-based maps (hardmax, sparsemax), the salience decay (ii) weakens τ_0's scored input while observed elements maintain or grow their salience, so the score gap is self-reinforcing. Provided the bias fluctuation $|b_{s+1}(\tau_0) - b_s(\tau_0)|$ does not compensate the salience loss, exact forgetting is irreversible. For positive maps ($\mathcal{A}_+$), exact forgetting cannot occur (Theorem 50).

Theorem 50 (Asymptotic forgetting; $\mathcal{A}_+$)**.** *Let $\varphi \in \mathcal{A}_+$ and suppose $|S(\tau)| \leq B_S$ for all τ and t. Then:*

(i) $a_t(\tau) > 0$ for μ-a.e. $\tau \in X$ and all $t \geq 0$. Exact forgetting does not occur.

(ii) For any measurable $A \subseteq X$ with $\mu(A) = \infty$ and any $\varepsilon > 0$, the set $\{\tau \in A : a_t(\tau) < \varepsilon\}$ has infinite measure for every t.

Under $\mathcal{A}_+$, forgetting is asymptotic: density approaches zero but never reaches it.

Proof. (i) By Theorem 21(iii) and Corollary 27, the scored input $S'_t(\tau) = h(S(\tau), \sigma_t(\tau), b_t(\tau))$ is bounded: $|S'_t(\tau)| \leq B$ for all τ, t. Since $\varphi \in \mathcal{A}_+$ assigns positive density to μ-a.e. τ whenever the input is bounded, $a_t(\tau) > 0$ μ-a.e.

(ii) The budget constraint $\int_X a_t\, d\mu = 1$ and $\mu(A) = \infty$ together imply that a_t cannot be bounded below on A by any $\varepsilon > 0$: otherwise $\int_A a_t\, d\mu \geq \varepsilon\, \mu(A) = \infty$, contradicting $\int_X a_t\, d\mu = 1$. Hence for each t, the set $\{\tau \in A : a_t(\tau) < \varepsilon\}$ has infinite measure. As t increases, the accumulator $C(\tau)$ grows only for elements with positive observation weight, so elements far from the active core receive negligible attention. □

Theorem 51 (Forced concentration; $\mathcal{A}$). *Let $\varphi \in \mathcal{A}$ and $\mu(X) = \infty$.*

(i) For any $\varepsilon > 0$ and any t, the set $\{\tau : a_t(\tau) \geq \varepsilon\}$ has μ-measure at most $1/\varepsilon$.

(ii) For any $\varepsilon > 0$ and any t, there exists a measurable set $K \subseteq X$ with $\mu(K) < \infty$ such that $\int_{X \setminus K} a_t \, d\mu < \varepsilon$.

The attention measure $a_t \, d\mu$ is essentially concentrated on a finite-measure set at every step.

Proof. (i) Markov's inequality: $\mu(\{a_t \geq \varepsilon\}) \leq \int_X a_t \, d\mu/\varepsilon = 1/\varepsilon$.

(ii) Let $X_1 \subset X_2 \subset \cdots$ be an exhaustion of X by finite-measure sets ($\bigcup X_n = X$, which exists since μ is σ-finite). Then $\int_{X_n} a_t \, d\mu \nearrow 1$ by monotone convergence. Choose n large enough that $\int_{X \setminus X_n} a_t \, d\mu < \varepsilon$ and set $K = X_n$. □

Theorem 52 (Path dependence of support; $\mathcal{A}$). *Let $\varphi \in \mathcal{A}$ (not necessarily positive). Two instances of the loop with identical structure (E, K, φ, λ) but different initial observation densities $a_0 \neq a_0'$ can produce active sets A_t and A_t' that become disjoint in finite time.*

Proof. Let $X = X_1 \sqcup X_2$ with $\mu(X_1), \mu(X_2) > 0$. Choose a_0 supported on X_1 and a_0' supported on X_2. Take φ to be hardmax (which is in $\mathcal{A}$ but not $\mathcal{A}_+$): $\varphi(f)$ concentrates mass on $\arg\max f$. Since $a_0(\tau) = 0$ for $\tau \in X_2$, the accumulators in X_2 never grow, so $\sigma(\tau) \to 0$ for $\tau \in X_2$ (Theorem 48(ii) applies once hardmax assigns zero to X_2). Meanwhile elements in X_1 maintain positive salience. For the hardmax, the argmax of $S' = S + \sigma + b$ remains in X_1 at every step, so $A_t \subseteq X_1$ for all $t \geq 0$. Symmetrically $A_t' \subseteq X_2$, giving $A_t \cap A_t' = \emptyset$ for all $t \geq 1$. □

Remark 53 (Finite collapse). In the finite case (μ counting measure on a finite set), all phenomena in this section collapse. Under $\mathcal{A}_+$, every element has positive weight at every step, the active set is always X, exact forgetting never occurs, and path dependence is limited to quantitative differences (which elements dominate), not qualitative ones (which elements survive). The infinite case reveals the budget constraint $\int a \, d\mu = 1$ as a topological phenomenon: the loop is forced to concentrate, forced to forget, and its trajectory depends irreversibly on initial conditions.

8 Open Problems

1. Does consolidation reduce the growth exponent of $|M|$ from $\Theta(t)$ to something sublinear, or only affect the constant factor?
2. Under what conditions does varying the query preserve distinct impressions indefinitely?
3. For $\varphi \in \mathcal{A} \setminus \mathcal{A}_+$, does a universal constant $\alpha > 0$ exist such that $a_t(\tau) \geq \alpha$ for all τ, t?
4. Are there results that require $\mathcal{A}_+$ but not exponential normalization?
5. What other mathematical facts about the loop follow from the existence of $|M^*|$?
6. What is the minimal controller specification needed to preserve the qualitative lifecycle?
7. Does the signal-control separation appear in other contexts?
8. Why does the loop emerge naturally when $|X| \gg |M^*|$?
9. Can an external controller that modulates $S(\tau)$ recover the density of asymptotically forgotten elements?
10. Is there a natural metric on attention histories under which $|M^*|$ is a critical threshold?
11. What happens when X grows over time or is nonstationary?

12. What qualitative insights emerge from systematic simulation?
13. How does the loop behave when inputs are non-deterministic?
14. What dynamics emerge when multiple loops interact?

A Accumulation Rigidity Details

The main-body proof of Proposition 7 gives a compact argument that $f(t) \sim t$ is necessary for $\sigma = C/N$ to converge to a nonzero constant. Here we give the full details: first the aggregate $\Theta(t^2)$ characterization of $\sum f$, then the two extremal cases that complete the necessity argument.

Throughout, C_t satisfies the generalized accumulation rule

$$C_{t+1} = C_t + a_t \cdot \frac{f(t)}{C_t},$$

where $f\colon \mathbb{N} \to \mathbb{R}_{>0}$ is a positive scaling function, $\delta \le a_t \le 1$ for some $\delta > 0$, $C_0 > 0$ (so $C_t > 0$ for all t by induction, since the increment is positive), and $N_t := N_0 + t$ with $N_0 \ge 1$. We write $\sigma_t = C_t/N_t$.

Proposition 54 (Aggregate accumulation rigidity; $\mathcal{A}$)**.** *If $\sigma_t \to \sigma_\infty \in (0,\infty)$, then $\sum_{s=0}^{t-1} f(s) = \Theta(t^2)$. Explicitly, there exist $\alpha, \beta > 0$ and $T \in \mathbb{N}$ such that for all $t \ge T$,*

$$\alpha\, N_t^2 \;\le\; \sum_{s=0}^{t-1} f(s) \;\le\; \beta\, N_t^2.$$

Proof. Upper bound on $\sum f$. Squaring the recurrence gives

$$C_{t+1}^2 = C_t^2 + 2a_t f(t) + \big(a_t f(t)/C_t\big)^2 \;\ge\; C_t^2 + 2\delta\, f(t),$$

since $a_t \ge \delta$ and $(a_t f(t)/C_t)^2 \ge 0$. Telescoping:

$$C_t^2 \;\ge\; C_0^2 + 2\delta \sum_{s=0}^{t-1} f(s). \tag{14}$$

From σ-convergence, $C_t \le 2\sigma_\infty N_t$ for large t. Combining with (14) and discarding $C_0^2 \ge 0$:

$$\sum_{s=0}^{t-1} f(s) \;\le\; \frac{2\sigma_\infty^2}{\delta}\, N_t^2.$$

Lower bound on $\sum f$. We first show $f(t) \le C_t^2$ eventually. From the recurrence, $a_t f(t) = C_t(C_{t+1} - C_t)$, so $f(t) \le C_t(C_{t+1} - C_t)/\delta$. Using σ-convergence with $\varepsilon = \sigma_\infty\delta/10$: for large t, $C_t = (\sigma_\infty \pm \varepsilon)N_t$ and $C_{t+1} - C_t \le (\sigma_\infty + \varepsilon) + 2\varepsilon N_t$, giving $f(t) = O(\varepsilon N_t^2)$. Since $C_t^2 \ge (\sigma_\infty - \varepsilon)^2 N_t^2$ and ε is small enough that $(\sigma_\infty - \varepsilon)^2\delta > 2\varepsilon(\sigma_\infty + \varepsilon)$, we get $f(t) \le C_t^2$ for large t.

Once $f(t) \le C_t^2$, the squared recurrence satisfies $C_{t+1}^2 \le C_t^2 + 3\, f(t)$ (since $(a_t f/C_t)^2 \le f^2/C_t^2 \le f$ and $2a_t f \le 2f$). Telescoping from T_0 (past which $f \le C^2$) and using $C_t \ge (\sigma_\infty/2)N_t$ for large t:

$$(\sigma_\infty/2)^2 N_t^2 \;\le\; C_t^2 \;\le\; C_{T_0}^2 + 3\sum_{s=T_0}^{t-1} f(s) \;\le\; C_{T_0}^2 + 3\sum_{s=0}^{t-1} f(s).$$

For N_t large enough that $(\sigma_\infty/2)^2 N_t^2 > 2\, C_{T_0}^2$:

$$\sum_{s=0}^{t-1} f(s) \;\ge\; \frac{(\sigma_\infty/2)^2}{6}\, N_t^2. \qquad \square$$

Proposition 55 (Superlinear f forces $\sigma \to \infty$; $\mathcal{A}$)**.** *If $f(t) \geq c\,(t+1)^2$ for some $c > 0$ and all t, then $\sigma_t \to \infty$.*

Proof. From (14), $C_t^2 \geq 2\delta c \sum_{s<t}(s+1)^2 \geq \delta c\, t^3/2$ (using $\sum_{s=0}^{t-1}(s+1)^2 \geq t^3/4$, verified by induction). For $t \geq N_0$, $N_t \leq 2t$, so

$$\sigma_t = \frac{C_t}{N_t} \;\geq\; \frac{\sqrt{\delta c/2}}{2}\sqrt{t} \;\to\; \infty. \qquad \square$$

Proposition 56 (Sublinear f forces $\sigma \to 0$; $\mathcal{A}$)**.** *If $f = o(t)$, then $\sigma_t \to 0$.*

Proof. Fix $\varepsilon > 0$; we show $\sigma_t \leq \varepsilon$ eventually. Set $\varepsilon' = \varepsilon/2$ and choose $\eta = \min(\varepsilon'^2/2,\ \varepsilon' C_0/2)$, so $\eta < \varepsilon'^2$. By sublinearity, $f(t) \leq \eta(t+1)$ for $t \geq T_1$.

Step 1: σ drops below ε'. Suppose $C_t \geq \varepsilon' N_t$ for all $t \geq T_1$. Each increment satisfies

$$C_{t+1} - C_t = \frac{a_t f(t)}{C_t} \;\leq\; \frac{\eta\, N_t}{\varepsilon'\, N_t} \;=\; \frac{\eta}{\varepsilon'},$$

so $C_{T_1+k} \leq C_{T_1} + k \cdot \eta/\varepsilon'$. But $C_{T_1+k} \geq \varepsilon' N_{T_1+k} = \varepsilon'(T_1 + k + N_0)$, giving $k(\varepsilon' - \eta/\varepsilon') \leq C_{T_1}$. Since $\eta < \varepsilon'^2$ implies $\varepsilon' - \eta/\varepsilon' > 0$, this fails for large k, a contradiction. So there exists $t_0 \geq T_1$ with $\sigma_{t_0} < \varepsilon'$.

Step 2: barrier. Set $M = \varepsilon' + \eta/C_0 \leq \varepsilon$ (using $\eta \leq \varepsilon' C_0/2$, so $\eta/C_0 \leq \varepsilon'/2$). We show $C_t \leq M N_t$ for all $t \geq t_0$ by induction. The base case holds since $C_{t_0} < \varepsilon' N_{t_0} < M N_{t_0}$. For the inductive step, assume $C_t \leq M N_t$ and consider two cases. If $C_t \geq \varepsilon' N_t$, the Step 1 bound gives $C_{t+1} - C_t \leq \eta/\varepsilon' \leq \varepsilon' \leq M$, so $C_{t+1} \leq M N_t + M = M N_{t+1}$. If $C_t < \varepsilon' N_t$, then since $C_0 \leq C_t$ (by monotonicity), $C_{t+1} - C_t \leq f(t)/C_0 \leq (\eta/C_0) N_t$, so $C_{t+1} \leq (\varepsilon' + \eta/C_0) N_t = M N_t \leq M N_{t+1}$. $\square$

B Formal Verification

The finite-domain case (X a finite type) of every proof in this paper has been formally verified in Lean 4 using the Mathlib library. The measure-theoretic results in Section 7 (support dynamics, exact and asymptotic forgetting, forced concentration, path dependence) apply to the infinite-domain generalization introduced in this paper and are not yet formalized.

Each Lean declaration is named to match its paper label.

For example, `accumulation_rigidity_aggregate` corresponds to Proposition 54.

The module structure mirrors the paper's sections:

- `Defs/`: core definitions (simplex maps, loop state, accumulation)
- `Core/`: main dynamics (convergence, contraction rate, Lyapunov, non-degeneracy)
- `Rigidity/`: rigidity theorems (accumulation, pipeline, capture, scoring, targets)
- `CaptureInertness/`: capture-consolidation cycle (Lipschitz, inertness, forced cycle)

To verify: install Lean 4 and Mathlib, then run `lake build` in the `AttentionLoop/` directory. The formalization and a GPU implementation in PyTorch are available at https://github.com/nemustudies/attention-loop/.

Notation

Symbol	Meaning
(X, Σ, μ)	σ-finite measure space
$V = \mathbb{R}^d$	Embedding space
$S\colon X \to \mathbb{R}$	Base score function (bounded, measurable)
φ	Simplex map: probability measure on (X, Σ), $\ll \mu$
$\lambda\colon [0,1] \to [0,1]$	Blend rate, $\lambda(0) = 1$, $\lambda(1) = 0$
$E\colon X \to V$	Embedding map (measurable, $R = \sup \lVert E(\tau) \rVert$ finite)
$K\colon V \to V$	Linear feedback map
M	State matrix (rows = stored impressions; always finite)
$a\colon X \to \mathbb{R}_{\geq 0}$	Observation density ($\int a \, d\mu = 1$)
$C\colon X \to \mathbb{R}_{>0}$	Accumulator (cumulative observation weight)
$N\colon X \to \mathbb{N}_{>0}$	Observation counter
$\sigma = C/N$	Salience (average observation weight)
$b\colon X \to \mathbb{R}$	Retrieval bias ($b = E^\top K \ell$)
$q \in V$	Context vector ($q = \int a \, E \, d\mu$)
$w \in \Delta^{\lvert M \rvert}$	Retrieval weights
$\ell \in V$	Retrieved impression
θ	Capture threshold
$R = \sup_{\tau \in X} \lVert E(\tau) \rVert$	Embedding radius (finite by assumption)
$A_t = \{\tau : a_t(\tau) > 0\}$	Active set (up to μ-null sets)
$\mathcal{P}(X)$	Probability measures on (X, Σ)
$\Delta^n = \{x \in \mathbb{R}^n_+ : \sum_i x_i = 1\}$	Probability simplex
$\mathcal{A}$	All simplex maps (density ≥ 0, $\int = 1$)
$\mathcal{A}_+ \subset \mathcal{A}$	Positive order-preserving simplex maps
$\mathcal{L}$	Set of admissible blend rates

Table 2: Summary of notation

A Finite-X Loop

When X is a finite set with counting measure, the infinite-dimensional loop simplifies to discrete sums. Let $X = \{1, \ldots, n\}$ be finite, $V = \mathbb{R}^d$, φ any simplex map, λ any blend rate, $E\colon X \to V$, and $K\colon V \to V$ linear. Initial conditions are $M_0 = \emptyset$, $C_0 > 0$, $N_0 = 1$.

Finite-X loop (counting measure)

X finite, $V = \mathbb{R}^d$, φ any simplex map, λ any blend rate, $E\colon X \to V$, $K\colon V \to V$ linear.
$M_0 = \emptyset$, $C_0 > 0$, $N_0 = 1$.

①	$a = \varphi(S + \sigma + b)$	**observe**
②	$C \leftarrow C + a \cdot (t{+}1)/C$ $N \leftarrow N + 1$ $\sigma = C/N$	**accumulate**
③	$q = aE$ $w = \varphi(q\, M^\top)$ $\ell = wM$ $b = EK\ell$	**retrieve**
④	$\lambda_j = \lambda(w_j / \max w)$ $M^* = \varphi\big((M{+}q)M^\top\big)M$ $M \leftarrow (1{-}\lambda)M + \lambda\, M^*$	**consolidate**
⑤	$\theta = \max(qM^\top)$ $M \leftarrow M \cup \{E(i) : q^\top E(i) > \theta\}$	**capture**
①	$a = \varphi(S + \sigma + b)$	**observe**

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The author is an undergraduate at the University of California, San Diego, has no background in formal mathematics or computer science, completed this work during their first year, and otherwise spends their time drawing anime girls as a digital artist. Contact: `nemustudies@gmail.com`.

A personal note

Hi. Thanks for reading this paper. I hope it gave you something new to explore, and that you discovered and understood something personal for yourself. I enjoyed sharing this with you, and working on this has taught me a lot about myself and about others. I don't know if I'll continue doing mathematics in this way after this, as I have other passions and hobbies (I want to get back to drawing, pass my classes, hang out with friends, take care of myself, and work on other things I think are interesting), but I felt like I understood a bit more why mathematicians love math so much.

I never thought I'd be up till 5 am every day working on math for 11 days straight while failing my classes, but here I am writing a math paper (and formalizing the whole thing with a proof calculator so that I could make sure it was real) since I wanted to share what I found. I did start getting sleep in the last third, though. This is my first time writing a paper and sharing it, so I appreciate you sticking with me with the quirks of the presentation.

Honestly, it was an accident that I even ended up fixated on finding this object. I feel very lucky. That being said, I had a lot of fun working on this and trying to discover a 0-degree-of-freedom solution to the problem. It was much more difficult than I expected.

I found a lot of interesting things and connections to things outside of math as I explored this object for myself, but I want to let you discover what these things personally mean to you, so I've chosen to keep the interpretation minimal. There's so much in the world to explore and see, and so many things to do and experience, and I hope that by reading this, you gain a new perspective.

I'm happy to be the medium through which this loop is shown to you. I hope you find it as exciting to explore as I did.

Once again, thank you for your time, and I'm thankful for the opportunity to share something so pretty with you.

—*nemu*